Title page and declarations

Manuscript title

Calibration of Heart-Rate Reserve for Graded Cycle Ergometry: An Endpoint-Preserving Transfer Evaluated Against Linear Alternatives

Running title

Endpoint-preserving HRR calibration

Author

BoTao Cai

Affiliation

College of Physical Education, Jimei University, Xiamen, Fujian, China

Corresponding author

BoTao Cai

Email: xcai2004xcai@gmail.com

ORCID: 0009-0002-3662-4539

Article type

Original research / methodological validation study

Word and item counts

- English abstract: 299 words.
- English main manuscript: approximately 4,840 words excluding references, tables, and legends.
- Main figures: 3.
- Main tables: 4.
- References: 14.

Author contributions

BoTao Cai: conceptualization, methodology, software, formal analysis, investigation, data curation, visualization, writing of the original draft, review and editing, and project administration.

Funding

This research received no external funding.

Competing interests

The author declares no competing interests.

Ethics statement

No new participants were recruited and no new human data were collected. The study analyzed public, deidentified datasets whose original records report ethics approval and consent. A separate institutional determination for this secondary analysis was not obtained. The author will provide an institutional exemption or not-human-participant-research determination if required by the target journal.

Data availability

The development dataset is available from Zenodo at https://doi.org/10.5281/zenodo.10841412. The ACTES external dataset is available from PhysioNet at https://doi.org/10.13026/2qs3-kh43. Processed source data are supplied in the accompanying workbook.

Code availability

The complete reproducible analysis is available at https://github.com/xcai66/tHRR-I-PMData. A new version-specific archival DOI will be inserted after the revised release is deposited by Zenodo.

Acknowledgements

The author thanks the creators of the Graded Incremental Test Data and ACTES datasets for making deidentified exercise-test data openly available.

Generative AI disclosure

During manuscript preparation, generative AI was used for language editing, document organization, and code assistance. The author verified the sources, analyses, and final manuscript and accepts full responsibility for the work. This wording should be adapted to the selected journal's policy.